POWER BI DATA TRANSFORMATION & LOADING

1. Introduction to Power Query Editor

Power Query Editor is the dedicated interface in Power BI for performing **data extraction, cleaning, transformation, formatting, and shaping** before loading the data into Power BI Desktop for visualization. It follows the **ETL model**:

- **Extract:** Connect to a data source (Excel, CSV, SQL, ERP exports, etc.)
- **Transform:** Clean, modify, reshape, and prepare the data for modeling
- **Load:** Apply transformations and load the final dataset to Power BI desktop

Power Query uses a scripting language called **M Language** internally, which automatically records each transformation step.

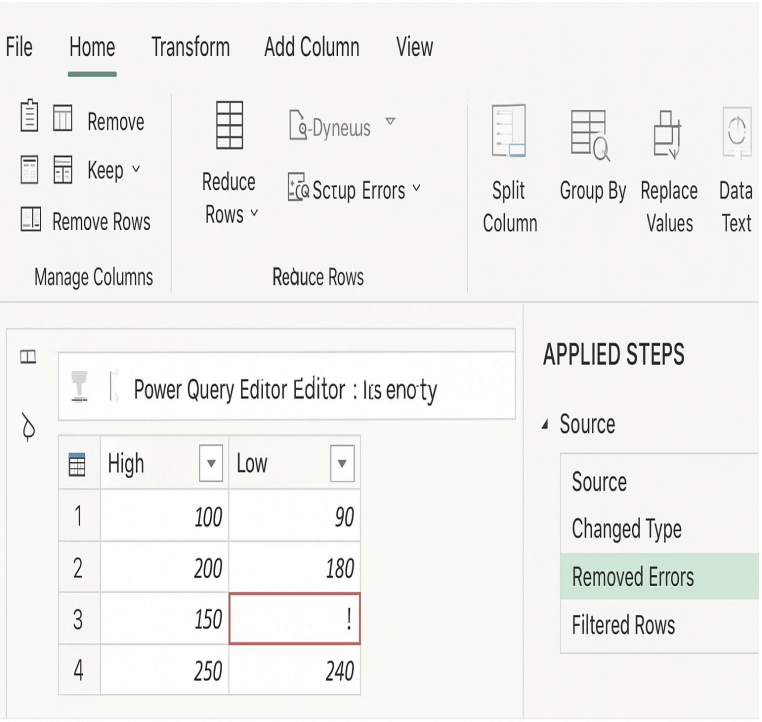
YATICORP
EMPOWERING EDUCATION

2. User-Friendly Interface & Logical Workflows

Power Query Editor is designed to be extremely **intuitive, user-friendly, and logical**. Most tasks follow a step-by-step approach, where each transformation automatically appears in the **APPLIED STEPS** pane.

Why It Is Easy to Use

- Ribbon-based layout similar to Excel
- Each option is descriptive and easy to understand
- Transformations are automatically recorded
- Errors are highlighted for quick correction



Ribbon-based layout comsssto day tos to be extremely intuitive, user-friendly

- Both row-level and column-level operations are available
- Undoing a step is as simple as deleting the applied step

3. Similarities Between the Transform and Add Column Tabs

Both the **Transform** and **Add Column** tabs contain options that look similar at a glance, because both involve operations on columns. The difference lies in the **purpose**:

Common Functionalities in Both Tabs

Both tabs include options like:

- **ABC (text transformations)**
- **Extract**
- **Parse**
- **Statistics**
- **Standard**
- **Scientific**
- **Trigonometry**
- **Rounding**
- **Information**
- **Date and Time**
- **Duration**

YATICORP
EMPOWERING EDUCATION

These operations allow manipulation, extraction, data type changes, and mathematical calculations directly on selected columns.

4. Differences Between Transform Tab and Add Column Tab

Although both appear similar, their functions serve **different goals**.

4.1 Transform Tab (Modifies Existing Columns)

The Transform tab **changes the current column permanently**. It does **not** create new columns.

Unique options available only in the Transform tab

- **Group By**
- **Transpose**
- **Pivot Column**
- **Unpivot Columns**
- **Reverse Rows**
- **Count Rows**
- **Convert to List**
- **Advanced statistical transformations**
- **R/Python script execution support**

YATICORP
EMPOWERING EDUCATION

Transform tab is ideal for:

- Reshaping tables
- Modifying existing data
- Aggregations
- Structure-level changes

4.2 Add Column Tab (Creates New Columns Without Altering Existing Data)

This tab is used to **retain the original data** and derive new calculated columns.

Features exclusive to Add Column:

- **Custom Column**
- **Invoke Custom Function**
- **Duplicate Column**
- **Index Column**
- **Conditional Column**
- **Extract values into a new column**

Unlike Transform, Add Column does **not** support:

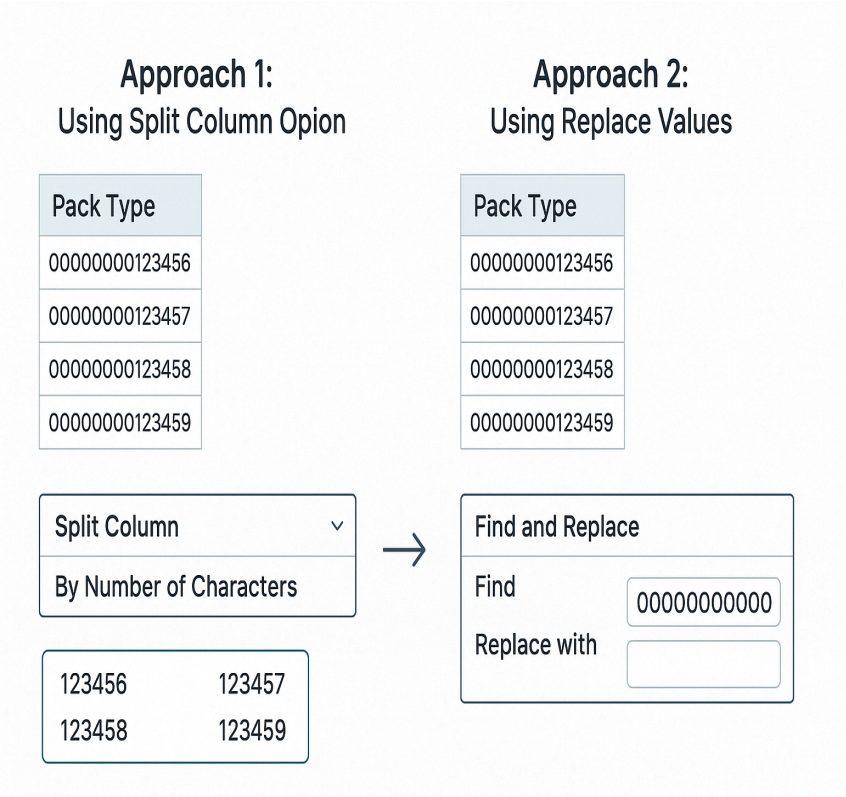
- R or Python integration
- Pivot/unpivot or group-by operations

YATICORP
EMPOWERING EDUCATION

5. Example Scenario: Extracting Only the Last Six Digits of a Pack Type

Assume a column "Pack Type" contains long numeric strings with leading zeros. If the requirement is to extract only the last 6 digits, there are two approaches:

Approach 1: Using Split Column Option



- Select column → Split Column → By Number of Characters
- Specify split from the left to remove unwanted digits
- Retain the last part

Approach 2: Using Replace Values

- Find the leading “000000000000” (12 zeros)
- Replace them with blank
- Result displays only the last six digits

This demonstrates the flexibility of Power Query: the same output can be achieved through multiple methods.

---

6. Data Preparation and Cleaning Process – Complete Example

YATICORP
EMPOWERING EDUCATION

The following section summarizes the end-to-end transformation steps performed during the practice exercise.

6.1 Connecting to a Data Source

- In Power BI Desktop → Home → **Get Data → Excel**
- Select file (ERP data file)
- Load required tables (e.g., "Sales Summary Report")

This creates the first step:
Source (loading the file)

6.2 Navigation to Required Table

Power Query navigates inside the Excel file structure to reach the correct table.

Step created:
Navigation
Shows which sheet/table was selected.

6.3 Automatic Data Type Detection

Power BI automatically reviews the values in each column and assigns a suitable data type such as:

- ABC (text)
- 123 (whole number)
- 1.2 (decimal number)
- Calendar icon (date)
- Time icon (time duration)

YATICORP
EMPOWERING EDUCATION

Although automatic detection is intelligent, data should still be manually verified, especially when:

- Data contains mixed formats
- Blank/NULL values exist
- A column logically needs a different type

Step created:
Changed Type

6.4 Removing Unnecessary Columns

Columns containing only NULL values (e.g., column 5, 8, 11) are removed.

Step created:
Removed Columns

6.5 Removing Top Rows

To remove unwanted headers or blank rows:

- Home tab → Remove Rows → Remove Top Rows
- Example: Removed first 5 rows
  Creates M function: Table.Skip(Source, 5)

Step created:
Removed Top Rows

6.6 Promoting Headers

Once the top unnecessary rows are deleted:

- Home → Use First Row as Headers

Step created:
Promoted Headers

YATICORP
EMPOWERING EDUCATION

6.7 Renaming Columns

Columns are renamed to user-friendly names:
Outlet, **Category**, **Pack Type**, **Item**, etc.

Step created:
Renamed Columns

6.8 Filtering Out Unwanted Rows

Example: Remove “Total” rows which interfere with calculations.

- Select column → Filter → Remove "Total"

Step created:
Filtered Rows

6.9 Filling Down Missing Values

To replace NULL values with the value above:

- Transform → Fill → Down

Useful when:

- Table is grouped by categories
- Sub-headers exist
- Data has hierarchical structures

Step created:
Filled Down

6.10 Creating New Calculated Columns

Using **Add Column → Custom Column** to compute:

Revenue per Quantity
= Gross Revenue / Sales Quantity

YATICORP
EMPOWERING EDUCATION

Step created:
Added Custom Column

If some rows have Sales Quantity = 0, this results in errors.

6.11 Removing Errors or Zero Quantity Rows

Remove rows where:

- Quantity = 0
- Custom column shows error

Step created:
Removed Errors

6.12 Changing Data Type for New Columns

The new measure (Revenue per Quantity) is converted into **Currency** type.

Step created:
Changed Type

6.13 Capitalizing Outlet Names

To ensure uniform formatting:

- Transform → Format → Capitalize Each Word

Step created:
Capitalized Each Word

6.14 Removing Unwanted Digits from Pack Type

Using Replace Values or Split Column removes unwanted 12 leading zeros.

Step created:
Replaced Value or **Split Column**

YATICORP
EMPOWERING EDUCATION

---

7. Applying and Loading Data

After all transformations:

1. Click **Close & Apply**
2. Power BI:
    - Processes all transformations
    - Executes the M script
    - Loads prepared data into Power BI Desktop

The final dataset appears in the **Fields Pane**.

This completes the **ETL cycle** inside Power BI

8. Good Practices While Working in Power BI

- Save your work frequently
- Avoid keeping large datasets open
- Close heavy applications to avoid memory overload
- Keep transformations modular
- Use consistent column names
- Validate data types before modeling

---

9. Automating Transformations for New ERP Downloads

One of the greatest strengths of Power Query is **repeatability and automation**.

Once all steps are recorded, any new file with the same structure can be refreshed automatically.

YATICORP
EMPOWERING EDUCATION

9.1 Creating a New Version of ERP Data

- Copy old Excel file
- Rename as ERP Data Updated
- Change some values for demonstration
- Save and close

9.2 Changing Data Source from Power Query Editor

- Home → Transform Data
- In Power Query → **Data Source Settings**
- Select the source → Change Source
- Browse for the new file (ERP Data Updated)

- Click OK
- Refresh Preview

All previous steps (skip rows, rename, filter, calculate, replace, etc.) are automatically applied to the new dataset.

This proves that **Power Query automatically replays all transformations** like a macro.

9.3 Changing Source from Power BI Desktop

Alternatively:

- Home → Data Source Settings
- Select source
- Click “Change Source”
- Choose Excel file
- Click OK → Refresh

YATICORP
EMPOWERING EDUCATION

Again, the same transformations are applied automatically.

---

10. Understanding Basic vs Advanced Source Modes

Basic Mode

- Used for simple file replacements
- Only the file path changes
- Best for standard ERP outputs

Advanced Mode

Allows configuring:

- Multiple file paths

- Folder structures
- Parameterized paths
- M-code level data sources

Used in large enterprise projects with:

- Multiple files
- Dynamic parameters
- Shared datasets

---

11. Privacy and Permissions Settings

Power BI allows setting data privacy levels:

- **Public**
- **Organizational**
- **Private**

YATICORP
EMPOWERING EDUCATION

You may also:

- Edit or clear permissions
- Manage gateway connections

These settings matter for:

- Security
- Compliance
- Enterprise workflows

---

12. Exporting Data Source Files (.PBIDS)

Power BI supports exporting data source configurations as:

.PBIDS – Power BI Data Source File

Useful when:

- Sharing data connection setup
- Enforcing uniform source paths across teams

13. Final Loading & Visualization

Once the new data source is refreshed:

- Close & Apply
- Power Query executes all steps
- Prepared data loads into Power BI Desktop

YATICORP
EMPOWERING EDUCATION

Now data is ready for:

- Visualizations
- Dashboards
- Reports
- Analytics
- Publishing and collaboration

The file is saved with extension:
.PBIX (Power BI Extracted File)

---

14. Summary of Entire ETL Process

1. **Connect to data source (Excel → ERP Data)**
2. **Navigation** to required sheet or table
3. **Verify/Assign Data Types**
4. **Remove Null Columns**
5. **Remove Top Rows**
6. **Promote Headers**
7. **Rename Columns**
8. **Filter Records** (e.g., remove "Total")
9. **Fill Down Missing Values**
10. **Create Custom Column** (Revenue per Quantity)
11. **Remove Errors** (0 quantity rows)
12. **Change Column Types**
13. **Format Text** (Capitalize Each Word)
14. **Clean Pack Type Values**
15. **Apply Changes (Close & Apply)**
16. **Load Prepared Data into Power BI Desktop**
17. **Save File (.PBIX)**
18. **Automate Transformations by Changing Source**
19. **Refresh Data and Continue Analysis**

YATICORP
EMPOWERING EDUCATION

15.Conclusion

Power Query Editor is a powerful, intuitive, and comprehensive platform within Power BI that simplifies data preparation and transformation. With its step-based architecture, automation capability, M-language recording, and user-friendly interface, Power Query enables both beginners and advanced users to efficiently prepare data for analysis and reporting. Once transformations are created, they can be reused infinitely simply by changing the data source—making Power BI an ideal tool for ETL processes in business analytics.

YATICORP
EMPOWERING EDUCATION